Boosting Sales at A.R Furniture through Strategic Inventory Optimization and Customer-Centric Design

Final report for the BDM capstone Project

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1. Executive Summary:

A.R. Furniture currently relies on manual inventory tracking and informal replenishment decisions, leading to inconsistent stock levels and limited SKU-level visibility. This project converts four months of sales data (April–July 2025) into a structured, risk-adjusted inventory decision framework.

The analysis combined traditional ABC classification, a multi-criteria composite ABC model, demand variability measurement using the Coefficient of Variation (CV), 3-month moving average trend tagging, and service-level-based safety stock computation.

Traditional ABC showed moderate profit concentration, with seven SKUs contributing approximately 75.8% of total profit. However, profit-only classification proved insufficient, as several high-value SKUs exhibited high demand variability and intermittent sales patterns.

To address this, a composite ABC model incorporated normalized profit (40%), sales frequency (30%), and demand stability (30%). This reclassified nine SKUs as Class A, reflecting operational importance beyond financial contribution alone.

Demand analysis revealed structural volatility across the portfolio: most SKUs displayed CV > 1 or multiple zero-sale months, with only one SKU classified as statistically stable. Safety stock evaluation further showed that 60–75% of reorder levels for several SKUs consisted of buffer stock rather than cycle stock. For example, Teak Sawn Timber carries 30 units of safety stock against an average monthly demand of 13.08.

These findings indicate that working capital exposure is driven primarily by demand variability rather than pure revenue concentration.

A differentiated replenishment strategy is therefore recommended, including continuous review for volatile Class A SKUs, periodic review for moderate items, and controlled buffer reduction for declining SKUs. This structured approach enables improved service reliability while reducing unnecessary capital lock-in.

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2. Detailed Explanation of Analysis Process and Methodology

2.1 Analytical Framework and Rationale

The analytical process was designed to move beyond profit-only classification toward a risk-adjusted inventory prioritization model.

Traditional inventory classification focuses primarily on financial contribution. However, inventory planning decisions also depend on demand stability, sales regularity, and short-term direction of movement. Therefore, this study adopted a structured, multi-stage framework integrating financial, statistical, and operational indicators.

The methodology consisted of:

- Data aggregation and validation
- Demand variability measurement
- Traditional ABC classification
- Multi-Criteria ABC modelling
- Demand trend classification and policy mapping

This layered approach ensures that inventory decisions reflect both profitability and operational risk.

2.2 Data Aggregation and Preparation

Transaction-level data (April–July 2025) were aggregated at SKU × Month level to create time-series demand profiles.

Monthly aggregation was necessary because:

- Variability metrics require periodic demand observations.
- Moving averages require consistent time intervals.
- Stability analysis requires comparable monthly demand values.

The preprocessing stage ensured numeric consistency and removed structural data inconsistencies to prevent distortion in standard deviation and normalization calculations.

This step improves analytical reliability but does not influence classification logic.

2.3 Demand Variability Measurement (Coefficient of Variation)

2.3.1 Justification for Using CV

Inventory control decisions depend heavily on demand predictability. Items with unstable demand require higher safety stock and tighter monitoring.

Standard deviation alone is insufficient because it measures absolute variability and is sensitive to demand scale. Therefore, the Coefficient of Variation (CV) was used as a scale-independent measure of relative variability.

$$CV = \frac{\sigma}{\mu}$$

Where:

- σ = Standard deviation of monthly demand
- μ = Average monthly demand

CV allows comparison across high-volume and low-volume SKUs, making it suitable for multi-item inventory analysis.

2.3.2 Operational Interpretation

- CV < 0.5 → Stable demand
- 0.5–1.0 → Moderate variability
- CV > 1 → Highly volatile / intermittent demand

Items with CV greater than 1 exhibit variability exceeding average demand, indicating elevated forecasting uncertainty.

Using CV aligns the model with established inventory management literature and risk-based stocking principles.

2.4 Traditional ABC Classification

2.4.1 Rationale

Traditional ABC classification is grounded in Pareto's principle, which states that a small proportion of items typically contribute the majority of value.

This method prioritizes SKUs based on total profit contribution.

Classification thresholds (70–20–10) are commonly used in operations management and reflect practical segmentation standards.

2.4.2 Method

Traditional ABC classification was performed based on Total Profit Contribution.

Steps:

- SKUs ranked in descending order of Total Profit.
- Profit Contribution calculated:

$$\text{Profit Contribution}_i = \frac{\text{Total Profit}_i}{\sum \text{Total Profit}}$$

- Cumulative percentage computed.
- Classification applied:
  - Class A → Top ~70% cumulative contribution
  - Class B → Next ~20%
  - Class C → Remaining

2.4.3 Limitation

While effective for financial prioritization, this method does not differentiate between stable and volatile items. As a result, high-profit but erratic SKUs may be over-prioritized.

2.5 Multi-Criteria ABC Classification

2.5.1 Rationale for Model Extension

To address limitations of single-criterion classification, a composite ABC Score was developed integrating:

- Financial impact (Profit)
- Demand regularity (Sales Frequency)
- Demand predictability (Stability derived from CV)

This ensures classification reflects both economic contribution and operational reliability.

2.5.2 Justification for Feature Selection

- Profit reflects revenue importance.

- Sales frequency reflects transaction regularity and demand continuity.
- Stability reflects predictability and safety stock sensitivity.

Together, these metrics provide a balanced representation of SKU importance.

Sales Frequency was selected instead of total quantity because quantity alone may represent bulk transactions rather than consistent demand. Frequency captures how regularly a SKU is sold, which is operationally more relevant for replenishment planning.

Demand Stability was derived from the Coefficient of Variation (CV), which measures variability relative to average demand. CV is preferred over standard deviation because it allows comparison across SKUs with different demand scales.

2.5.3 Normalization

Min–Max normalization was applied:

$$X_{norm} = \frac{X_i - Min(X)}{Max(X) - Min(X)}$$

This standardizes all features between 0 and 1.

2.5.4 Normalization Justification

Because metrics operate on different scales (profit in currency, frequency in counts, CV as ratio), Min–Max normalization was applied to ensure comparability and prevent scale dominance.

Demand stability was obtained by inverting normalized CV:

$$Stability = 1 - CV_{normalized}$$

This transformation ensures consistent directional interpretation: higher score indicates better performance.

2.5.6 Composite ABC Score

$$ABC\ Score = 0.4 \times Normalized\ Profit + 0.3 \times Normalized\ Sales\ Frequency + 0.3 \times Normalized\ Stability$$

Profit was weighted higher (40%) due to direct financial impact, while frequency and stability were each weighted 30% to reflect operational reliability.

2.5.7 Weighting Rationale

The weighting scheme (0.4 / 0.3 / 0.3) reflects:

- Slightly higher emphasis on financial contribution
- Equal importance to operational reliability and regularity

This balanced weighting avoids over-dominance of any single factor.

2.5.8 Classification Threshold Justification

A 70% cumulative threshold was retained to maintain comparability with traditional ABC standards while reflecting composite importance.

This ensures operational alignment without distorting industry classification norms.

2.6 Demand Trend Analysis

2.6.1 Objective

Trend tagging was performed to convert static ABC classification into actionable inventory rules.

2.6.2 Method

Two moving averages were computed:

$$SMA_1 = \text{Average (Apr, May, Jun)}$$
$$SMA_2 = \text{Average (May, Jun, Jul)}$$

Trend determined using thresholds:

- $SMA_2 > 1.10 \times SMA_1 \rightarrow$ Rising
- $SMA_2 < 0.90 \times SMA_1 \rightarrow$ Declining
- Within ±10% → Stable

A ±10% threshold was selected to filter minor noise while capturing material demand shifts.

2.6.3 Volatility Override

If:

- CV > 1

  or
- ≥ 2 zero-sale months

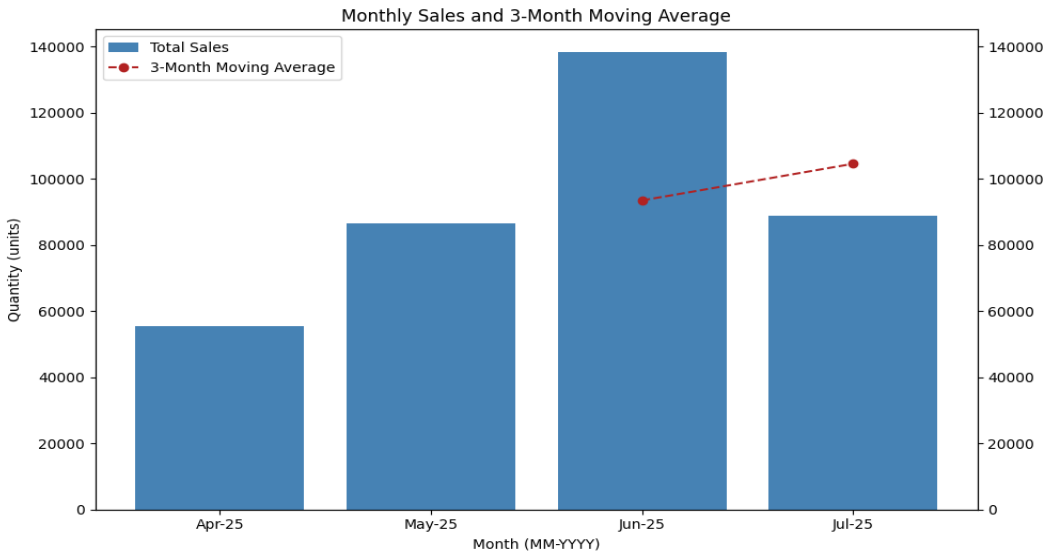
Then SKU classified as **Unpredictable**, overriding trend classification.

This rule is consistent with intermittent demand literature (Syntetos & Boylan, 2005).

3. Results and Findings

3.1 Overall Sales Performance (April–July 2025)

Figure 3.1 Monthly Sales and 3-Month Moving Average



The monthly sales trend indicates progressive growth from April to June 2025, followed by marginal stabilization in July. The peak observed in June is primarily driven by significant volume movement in Teak Sawn Timber and select frame components, rather than uniform growth across all SKUs.

The three-month moving average confirms mid-period acceleration in demand, suggesting temporary demand expansion rather than sustained structural growth.

However, SKU-level inspection reveals that several items exhibit zero-sale months, indicating intermittent purchasing behaviour rather than continuous consumption patterns.

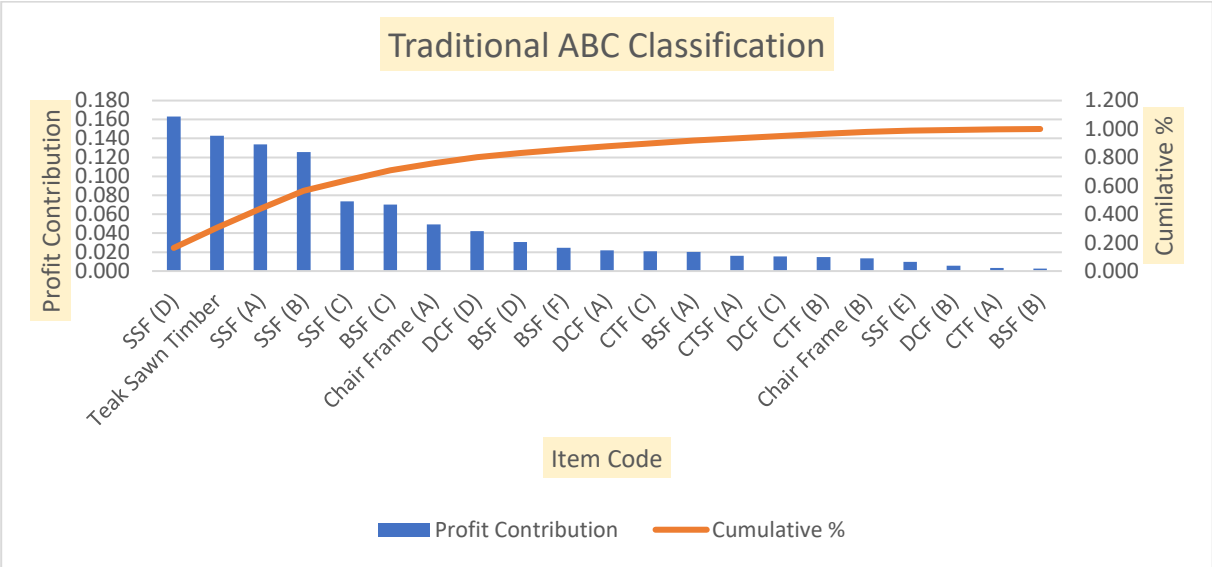
This uneven demand distribution justifies the need for structured classification and risk-adjusted inventory planning.

3.2 Traditional ABC Classification (Profit-Based)

Traditional ABC classification was conducted by ranking all SKUs in descending order of total profit contribution and computing cumulative contribution to overall profit. The 70–20–10 segmentation principle was applied to classify items into Classes A, B, and C.

The analysis reveals a moderately concentrated revenue structure. The highest contributing SKU, **SSF (D)**, accounts for approximately **16.3%** of total profit. The top two SKUs contribute **30.6%**, while the top four collectively account for **56.5%** of total profit. The cumulative contribution reaches **75.8%** at the seventh SKU (Chair Frame A), establishing the Class A boundary.

Figure 3.2 Traditional ABC Classification



3.2.1 Final classification:

- **Class A:** 7 SKUs (up to 75.8% cumulative)
- **Class B:** 5 SKUs (75.8%–89.8%)
- **Class C:** 9 SKUs (remaining 10.2%)

3.2.2 Key Findings:

- Profit concentration is strong but not extreme, the top 7 items generate approximately three-fourths of total profit.
- Individual contribution drops sharply after SSF (B), indicating diminishing marginal impact.
- Class C contains many SKUs with very low individual profit contribution (<2%).

3.2.3 Interpretation:

Traditional ABC clearly identifies revenue-dominant SKUs but does not incorporate demand stability, sales frequency, or risk exposure. The classification is purely financial and reflects revenue concentration structure.

3.3 Multi-Criteria ABC Classification (Composite Model)

To address the limitations of profit-only prioritization, a composite ABC model was developed integrating three normalized dimensions:

- Normalized Profit (financial importance)
- Normalized Sales Frequency (demand regularity)
- Normalized Stability (1 − normalized CV)

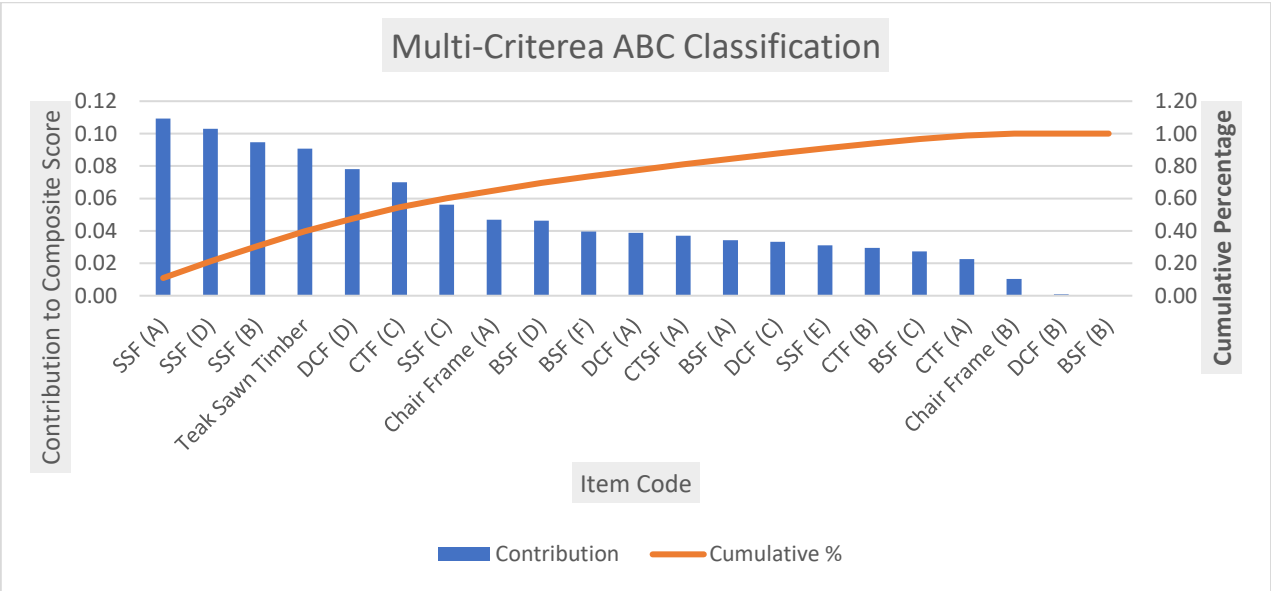
The composite score was calculated using the weighted formula:

ABC Score = 0.4 × Profit + 0.3 × Sales Frequency + 0.3 × Stability

SKUs were then ranked based on composite score contribution, and cumulative percentages were computed using the same 70% threshold principle for classification.

The cumulative contribution pattern differs meaningfully from traditional ABC. The rise in cumulative curve is more gradual, indicating a more balanced distribution of importance across top SKUs.

Figure 3.3 Multi-Criteria ABC Classification



3.3.1 Key findings include:

- SSF (A) alone contributes approximately 11% of total composite importance.
- Top 3 SKUs contribute 31%.
- Top 5 contribute 48%.
- Top 7 contribute 60%.
- 70% cumulative threshold is reached at BSF (D).
- 80% cumulative threshold is reached at CTSF (A).

3.3.2 Final composite classification:

- **Class A:** 9 SKUs (up to 70% cumulative composite score)
- **Class B:** 5 SKUs (70%–88%)
- **Class C:** 7 SKUs (remaining)

Compared to traditional ABC, the composite model produces a broader Class A group. This reflects operational balancing rather than strict financial dominance.

3.3.3 Several structural differences are observed:

- Some high-profit SKUs move lower in ranking due to elevated CV values, reducing their stability score.
- Stable items such as CTF (C) improve ranking despite moderate profit levels.
- SKUs with CV = 2 receive near-zero stability scores and fall into lower classes despite financial contribution.

3.3.4 Final Conclusion

The composite model therefore penalizes volatility and rewards demand consistency. It captures not only "how much profit an SKU generates," but also "how reliably it performs."

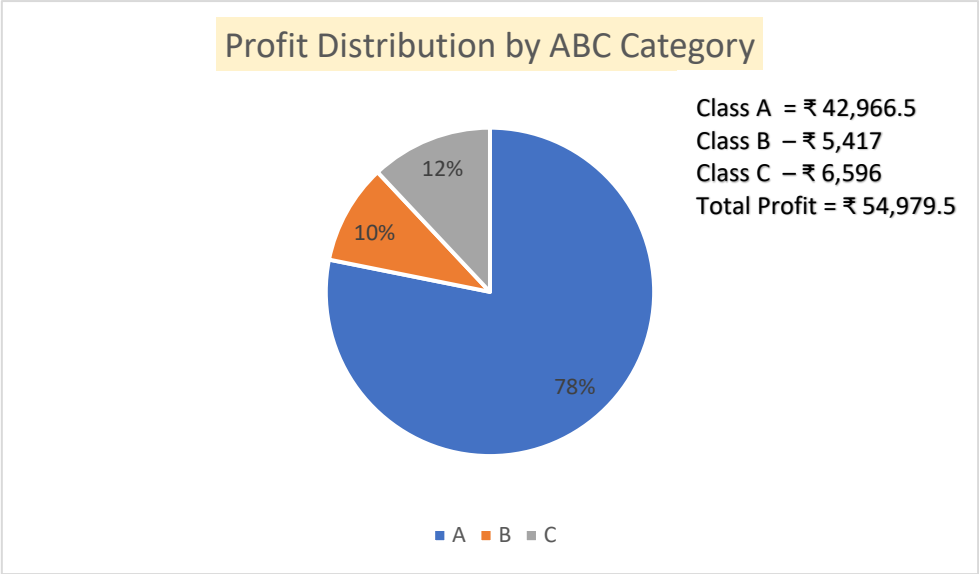
This results in a risk-adjusted prioritization structure that is more aligned with inventory control objectives. Unlike traditional ABC, which identifies revenue concentration, the multi-criteria model identifies economically important and operationally manageable SKUs.

The shift from 7 Class A SKUs (traditional) to 9 Class A SKUs (composite) indicates that operational reliability redistributes importance across items that may not be top profit generators but contribute consistently and predictably.

Overall, the multi-criteria approach provides a more robust foundation for replenishment planning, safety stock allocation, and capital efficiency management.

3.4 Profit Distribution by Multi-Criteria ABC Category

Figure 3.4 Profit Distribution Graph:



The profit contribution by ABC class is presented in Figure 3.2 (Pie Chart).

Class A items contribute approximately **78.15% of total profit**, despite representing only 9 out of 21 SKUs. Class B contributes about 9.85%, while Class C contributes 12%.

This confirms strong Pareto concentration within the inventory portfolio, where a limited number of SKUs generate the majority of financial value. The classification therefore justifies prioritised monitoring and tighter inventory control for Class A items.

3.5 Demand Trend Classification and Findings

3.5.1 High Demand Irregularity

The majority of SKUs were classified as Unpredictable, driven by:

- CV values greater than 1
- Multiple zero-sale months

This indicates intermittent and unstable demand across a significant portion of the inventory.

3.5.2 Limited Truly Stable Items

Only **CTF(C)** qualified as Stable (CV = 0.61 with consistent movement). This suggests that steady, low-variance demand is rare in the current SKU mix.

3.5.3 Identified Rising SKUs

- DCF (D)
- SSF (A)
- SSF (B)

This show sustained upward movement with acceptable volatility levels.

3.5.4 Identified Declining SKU

- SSF (D)

SMA_2 was meaningfully lower than SMA_1, indicating contraction.

3.5.5 Borderline Observations

Some SKUs appear directionally increasing but remain classified as Unpredictable due to high CV or multiple zero months.

This reflects short data horizon sensitivity and reinforces the need to prioritise statistical consistency over visual interpretation.

3.5.6 Overall Insight

Demand behaviour is structurally volatile, even among several Class A SKUs.

This implies that inventory control decisions cannot rely on profitability alone and must incorporate demand stability.

The revised classification provides a disciplined and reproducible approach to distinguishing:

- Directional growth
- True stability
- Intermittent unpredictability

3.6 Safety Stock and Reorder Level Insights

Safety stock was computed using service-level-based Z-values derived from the standard normal distribution. The Z-score represents the number of standard deviations required to achieve a specified

probability of avoiding stockout during lead time. In alignment with inventory criticality under ABC classification, differentiated service levels were applied: Class A items were assigned Z = 1.88 (≈97% service level), Class B items Z = 1.65 (≈95% service level), and Class C items Z = 1.28 (≈90% service level). This differentiation ensures that financially critical items receive higher protection against demand variability, while lower-impact items operate under leaner inventory policies. The approach reflects a risk-adjusted inventory control strategy, balancing service performance and working capital efficiency:

- Class A → Z = 1.88
- Class B → Z = 1.65
- Class C → Z = 1.28

Reorder Level = Average Demand + Safety Stock

3.6.1 Key Observations

3.6.1.1 Protective Buffer Dominance in Volatile SKUs

For several high-CV items, safety stock exceeds or closely matches average demand:

- BSF (C):

  Average = 1, Safety Stock = 3

- DCF (C):

  Average = 4, Safety Stock = 10

- Teak Sawn Timber:

  Average = 13.08, Safety Stock = 30

In such cases, the majority of inventory is buffer stock rather than cycle stock.

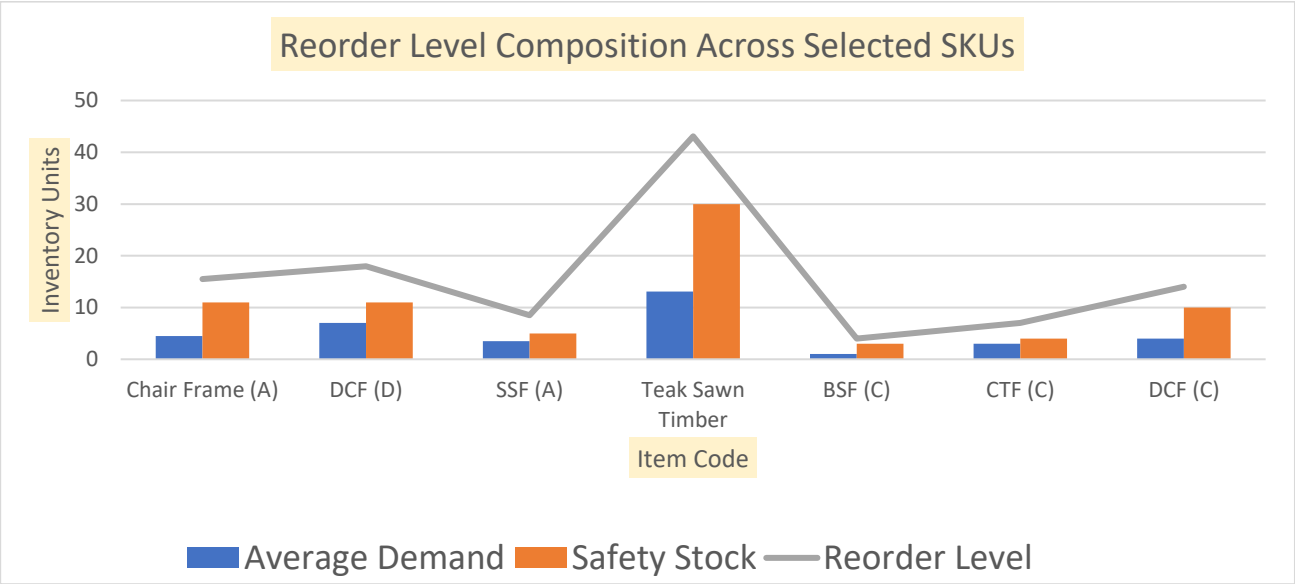
3.6.1.2 Working Capital Sensitivity in Class A Items

Class A SKUs carry significantly larger buffers due to higher service levels:

- DCF (D): Safety Stock = 11
- Chair Frame (A): Safety Stock = 11
- Teak Sawn Timber: Safety Stock = 30

This creates strong capital lock-in in volatile but critical items.

Figure 3.6 Reorder Level Composition Across Selected SKUs



3.6.1.3 Efficient Inventory in Stable SKUs

Low-CV items show relatively controlled buffers:

- CTF (C): Average = 3, Safety Stock = 4
- SSF (A): Average = 3.5, Safety Stock = 5

Here, inventory is more demand-aligned and less risk-driven.

3.6.2 Strategic Interpretation

The analysis reveals:

- High variability drives disproportionately high safety stock
- Class A volatility significantly influences capital allocation
- Intermittent SKUs require cautious replenishment strategies
- Stable SKUs allow leaner inventory positioning

Overall, inventory risk is structurally embedded within demand variability rather than solely financial ranking.

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4. Interpretation of Results and Recommendations

4.1 Integrated Inventory Risk Profile

The combined analysis reveals that inventory risk at A.R. Furniture is not driven solely by financial concentration, but by the interaction between profitability, demand variability, and trend behaviour.

Traditional ABC analysis confirms revenue concentration within a limited set of SKUs. However, the multi-criteria and variability analysis shows that several financially critical SKUs exhibit moderate to high demand uncertainty. This creates structural exposure where high-value items also require substantial safety buffers.

The demand trend classification further indicates that a significant portion of the portfolio is classified as "Unpredictable," reflecting intermittent or irregular monthly demand patterns. Only a limited number of SKUs demonstrate stable behaviour with low variability.

Thus, inventory exposure is multidimensional:

- Financial concentration risk
- Variability-driven operational risk
- Capital lock-in through safety stock

This confirms that a uniform replenishment policy would be suboptimal.

4.2 Strategic Segmentation of SKUs

To translate analytical findings into operational guidance, SKUs were segmented based on ABC class, variability (CV), trend, and safety stock ratio.

Table 4.2: Integrated SKU Risk Segmentation and Recommended Inventory Policy

SKU	ABC	CV	Trend	Risk	Safety Stock / Reorder Level	Strategic Action
Teak Sawn Timber	A	1.22	Unpredictable	Moderate	0.7	Continuous review; monitor variability closely; explore supplier lead-time reduction
DCF (D)	A	0.82	Rising	Moderate	0.61	Maintain high service level; increase forecast frequency; monitor growth sustainability
Chair Frame (A)	A	1.28	Unpredictable	High	0.71	Tight monitoring; consider demand smoothing; review safety stock quarterly
CTF (C)	A	0.61	Stable	Low	0.57	Lean replenishment; maintain service level; opportunity to gradually reduce buffer
SSF (D)	A	0.71	Declining	Low	0.6	Gradually reduce reorder level; prevent excess holding; review demand trend monthly
DCF (C)	B	1.41	Unpredictable	High	0.71	Periodic review system; avoid over-buffering; monitor intermittency
BSF (C)	C	2	Unpredictable	High	0.75	Lower service level acceptable; avoid capital lock-in; order cautiously

4.2.1 Critical and Volatile SKUs

Examples: Teak Sawn Timber, Chair Frame (A)

Characteristics:

- Class A financial importance
- CV > 1
- High safety stock ratios (~0.70)
- Unpredictable trend

Interpretation:

These items combine revenue criticality with demand uncertainty, resulting in significant capital allocation toward buffer stock.

Recommendation:

- Continuous review system
- Monthly forecast updates
- Supplier lead-time reduction initiatives
- Quarterly safety stock recalibration

4.2.2 Critical but Stable SKUs

Examples: CTF (C), SSF (A)

Characteristics:

- Class A
- CV < 0.8
- Lower safety stock ratio (~0.57–0.59)
- Stable or Rising trend

Interpretation:

These items are financially important yet operationally predictable.

Recommendation:

- Maintain high service level

- Gradually optimise safety stock downward
- Lean replenishment strategy

These SKUs offer opportunity for working capital efficiency without compromising service.

4.2.3 Moderate Value but High Variability SKUs

Examples: DCF (C), BSF (C)

Characteristics:

- Class B or C
- CV ≥ 1.4 (some at 2.0)
- High safety stock ratio (~0.75)
- Unpredictable trend

Interpretation:

Although financially less significant, these SKUs consume disproportionate safety buffers relative to their value contribution.

Recommendation:

- Periodic review system
- Lower service level tolerance
- Avoid excess stocking
- Consider minimum order or on-demand replenishment

This prevents capital being locked in low-impact volatile items.

4.2.4 Declining SKUs

Example: SSF (D)

Characteristics:

- Class A
- Low CV
- Declining trend

Interpretation:

Despite financial importance, declining momentum increases risk of overstock.

Recommendation:

- Gradual reduction in reorder level
- Conservative replenishment
- Monitor for structural demand shift

4.3 Working Capital Implications

The safety stock ratio analysis indicates that for several SKUs, more than 60–75% of reorder level consists of buffer stock rather than cycle stock.

This suggests that:

- Inventory structure is risk-driven rather than demand-driven.
- High variability directly inflates capital commitment. Class A items with moderate variability exert the strongest influence on working capital.
- Reducing demand uncertainty (via forecasting improvement or lead-time compression) would mathematically reduce safety stock requirements and improve capital efficiency.

4.4 Inventory Policy Recommendations

Based on integrated findings, the following policy framework is recommended:

4.4.1 Differentiated Replenishment System

- Continuous review (Q-system) for Class A volatile SKUs
- Periodic review (P-system) for Class B and C
- Reduced monitoring for stable low-risk SKUs

4.4.2 Dynamic Safety Stock Adjustment

- Increase monitoring frequency for Rising SKUs
- Reduce buffers gradually for Declining SKUs
- Recalculate CV quarterly

4.4.3 Lead-Time Reduction Strategy

- Since safety stock is proportional to variability and lead time:
- Reducing supplier lead time will directly reduce buffer requirement.
- Supplier negotiation and delivery scheduling should be explored.

4.4.4 Capital Efficiency Monitoring

Introduce KPI:

- Safety Stock / Average Demand ratio
- If ratio exceeds 2, conduct policy review.
- This institutionalises inventory discipline.

4.5 Strategic Conclusion

The integrated ABC, variability, trend, and safety stock analysis demonstrates that inventory optimisation at A.R. Furniture requires a risk-adjusted, differentiated control strategy.

Financial importance alone is insufficient for prioritisation. Demand uncertainty and trend direction significantly influence working capital exposure and service performance.

A structured, data-driven replenishment framework balancing service level and capital efficiency will enhance resilience and operational performance.